

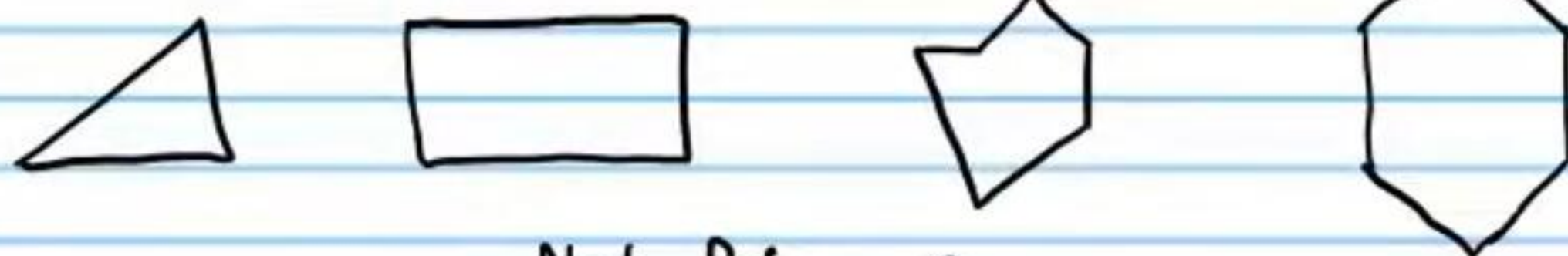
Special Quadrilaterals (Part I) and Angles of Polygons

Angles of Polygons

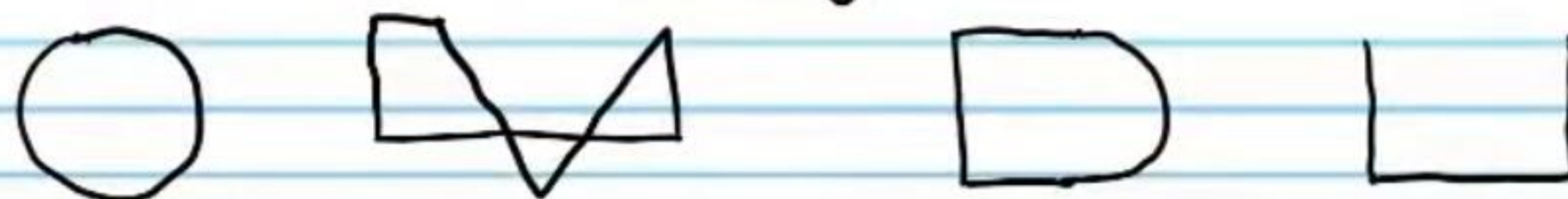
Review

Polygon: A closed plane figure whose sides are line segments that intersect only at endpoints.

Examples of Polygons



Not Polygons

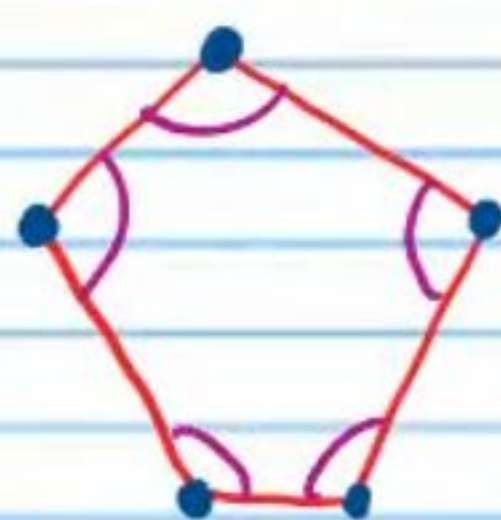


Parts of a Polygon

Vertices

Sides

Interior Angles



Names of Polygons

Sides

3
4
5
6
7

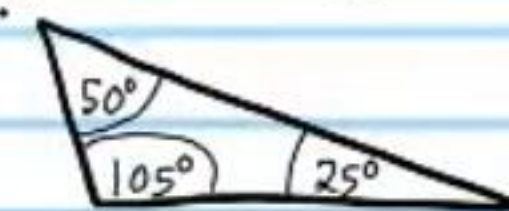
Name

Triangle
Quadrilateral
Pentagon
Hexagon
Heptagon

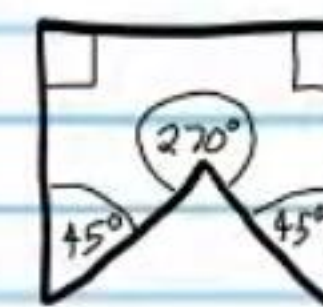
8
9
10
12
n

Octagon
Nonagon
Decagon
Dodecagon
n-gon

Convex Polygon: A polygon with no interior angle measures greater than 180° .



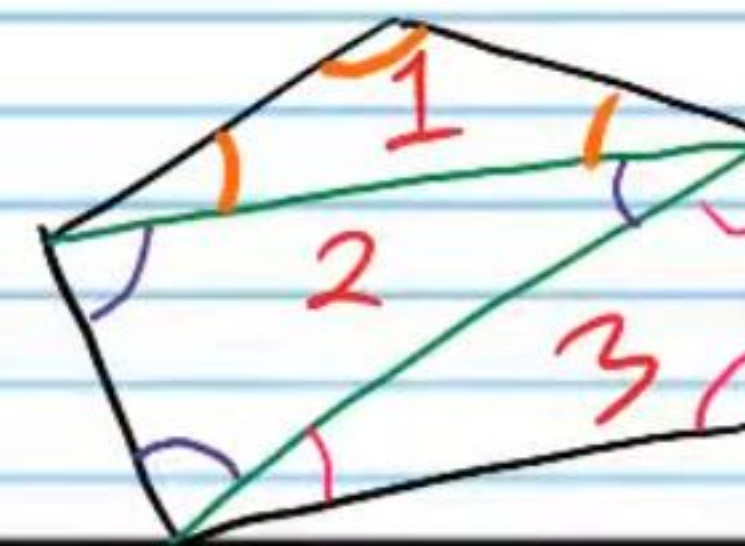
Concave Polygon: A polygon with one or more interior angles that measure greater than 180° .



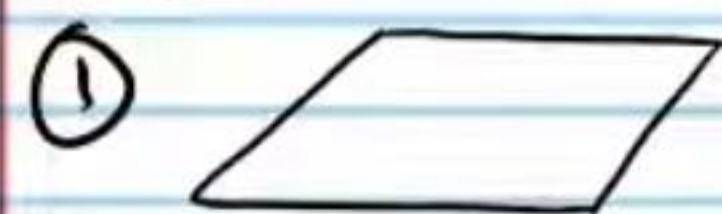
Polygon Angle Theorems

Polygon Interior Angles Theorem: If a convex polygon has n sides, then the sum of the measures of the interior angles is $180(n-2)$.

Explanation



Find the sum of the measures of the interior angles of the following polygons.

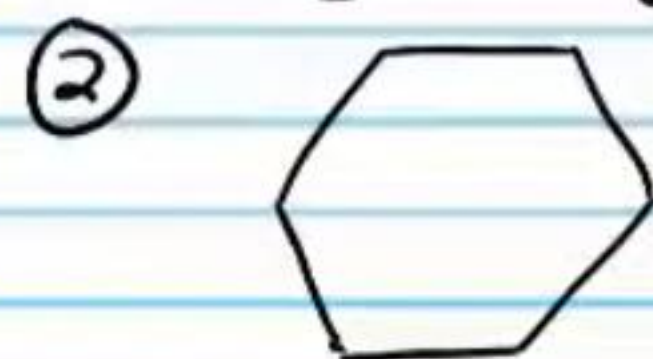


$$S = 180(n-2)$$

$$S = 180(4-2)$$

$$S = 180(2)$$

$$\boxed{S = 360^\circ}$$



$$S = 180(n-2)$$

$$S = 180(6-2)$$

$$S = 180(4)$$

$$\boxed{S = 720^\circ}$$

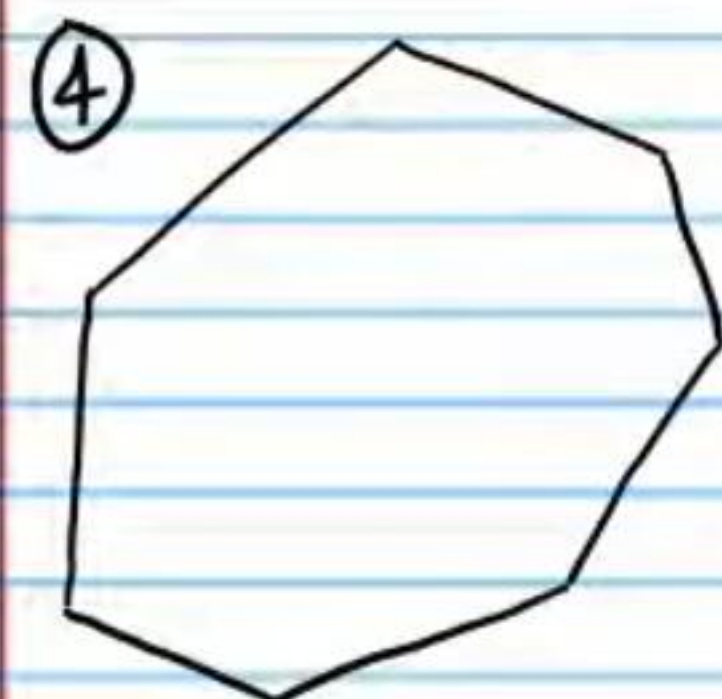


$$S = 180(n-2)$$

$$S = 180(8-2)$$

$$S = 180(6)$$

$$\boxed{S = 1,080^\circ}$$

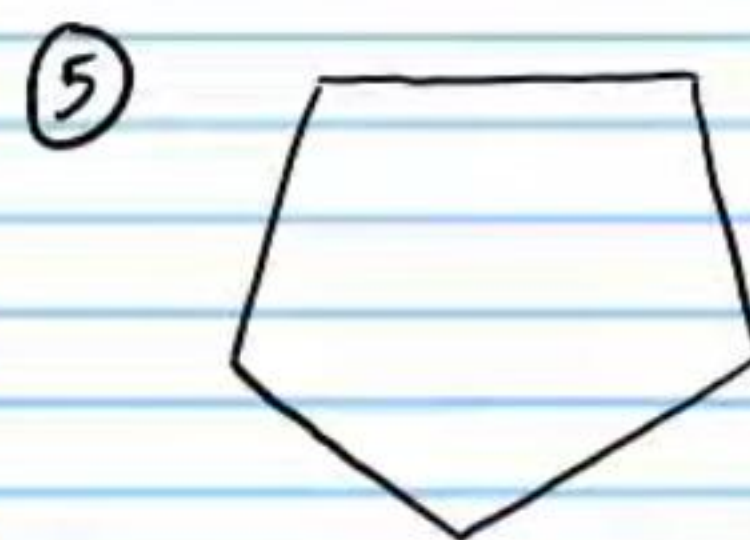


$$S = 180(n-2)$$

$$S = 180(7-2)$$

$$S = 180(5)$$

$$\boxed{S = 900^\circ}$$

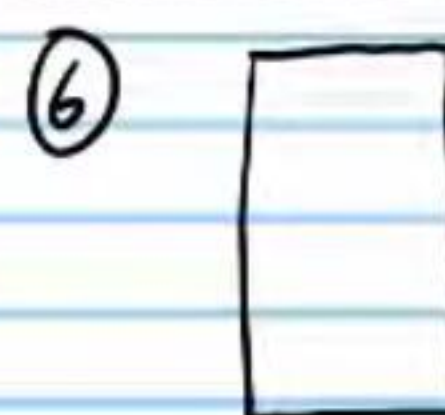


$$S = 180(n-2)$$

$$S = 180(5-2)$$

$$S = 180(3)$$

$$\boxed{S = 540^\circ}$$



$$S = 180(n-2)$$

$$S = 180(4-2)$$

$$S = 180(2)$$

$$\boxed{S = 360^\circ}$$

n-gon: A polygon with n sides.

Find the sum of the measures of the interior angles of the following convex polygons.

⑦ Decagon

$$S = 180(n-2)$$

$$S = 180(10-2)$$

$$S = 180(8)$$

$$\boxed{S = 1,440^\circ}$$

⑧ Heptagon

$$S = 180(n-2)$$

$$S = 180(7-2)$$

$$S = 180(5)$$

$$\boxed{S = 900^\circ}$$

⑨ 13-gon

$$S = 180(n-2)$$

$$S = 180(13-2)$$

$$S = 180(11)$$

$$\boxed{S = 1,980^\circ}$$

⑩ Dodecagon

$$S = 180(n-2)$$

$$S = 180(12-2)$$

$$S = 180(10)$$

$$\boxed{S = 1800^\circ}$$

⑪ Octagon

$$S = 180(n-2)$$

$$S = 180(8-2)$$

$$S = 180(6)$$

$$\boxed{S = 1,080^\circ}$$

⑫ 23-gon

$$S = 180(n-2)$$

$$S = 180(23-2)$$

$$S = 180(21)$$

$$\boxed{S = 3,780^\circ}$$

If each number below is the sum of the measures of the interior angles of a convex polygon, find the number of sides of each polygon.

⑬ $2,700^\circ$

$$S = 180(n-2)$$

$$2700 = 180(n-2)$$

$$2700 = 180n - 360$$

$$3,060 = 180n$$

$$\boxed{17 = n}$$

⑭ $3,780^\circ$

$$S = 180(n-2)$$

$$3780 = 180(n-2)$$

$$3780 = 180n - 360$$

$$4,140 = 180n$$

$$\boxed{23 = n}$$

⑮ $1,980^\circ$

$$S = 180(n-2)$$

$$1980 = 180(n-2)$$

$$1980 = 180n - 360$$

$$2,340 = 180n$$

$$\boxed{13 = n}$$

⑯ $4,500^\circ$

$$S = 180(n-2)$$

$$4500 = 180(n-2)$$

$$4500 = 180n - 360$$

$$4860 = 180n$$

$$\boxed{27 = n}$$

⑰ $2,880^\circ$

$$S = 180(n-2)$$

$$2880 = 180(n-2)$$

$$2880 = 180n - 360$$

$$3,240 = 180n$$

$$\boxed{18 = n}$$

⑱ $2,340^\circ$

$$S = 180(n-2)$$

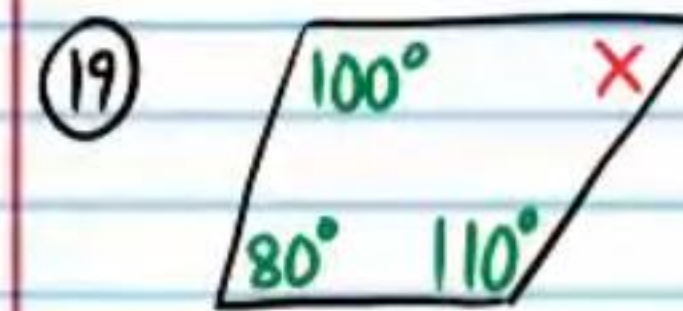
$$2340 = 180(n-2)$$

$$2340 = 180n - 360$$

$$2,700 = 180n$$

$$\boxed{15 = n}$$

Find the unknowns in each polygon below.



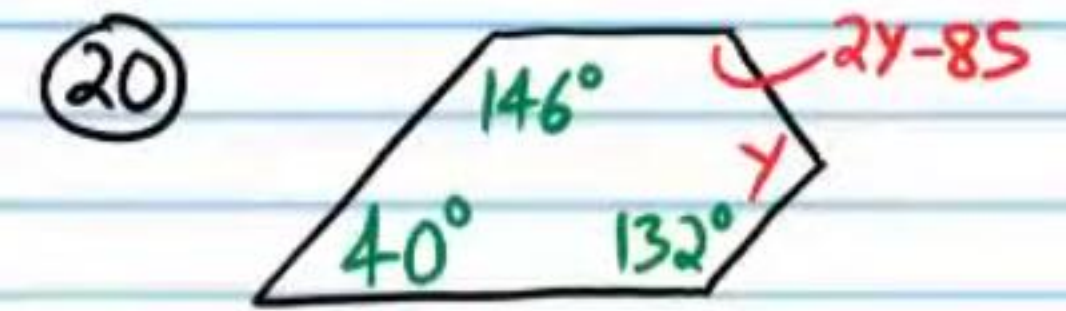
I) $S = 180(n-2)$

$$S = 180(4-2) = 360^\circ$$

II) $360 = 100 + 80 + 110 + x$

$$360 = 290 + x$$

$$\boxed{70^\circ = x}$$



I) $S = 180(n-2)$

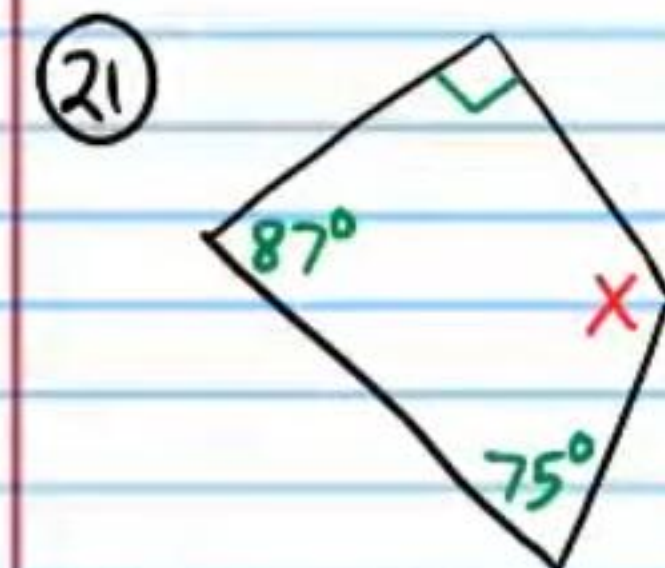
$$S = 180(5-2) = 540^\circ$$

II) $540 = 40 + 146 + 132 + y + 2y - 85$

$$540 = 233 + 3y$$

$$307 = 3y$$

$$\boxed{102.\bar{3}^\circ = y}$$



I) $S = 180(n-2)$

$$S = 180(4-2) = 360^\circ$$

II) $360 = 90 + 87 + 75 + x$

$$360 = 252 + x$$

$$\boxed{108^\circ = x}$$



I) $S = 180(n-2)$

$$S = 180(6-2) = 720^\circ$$

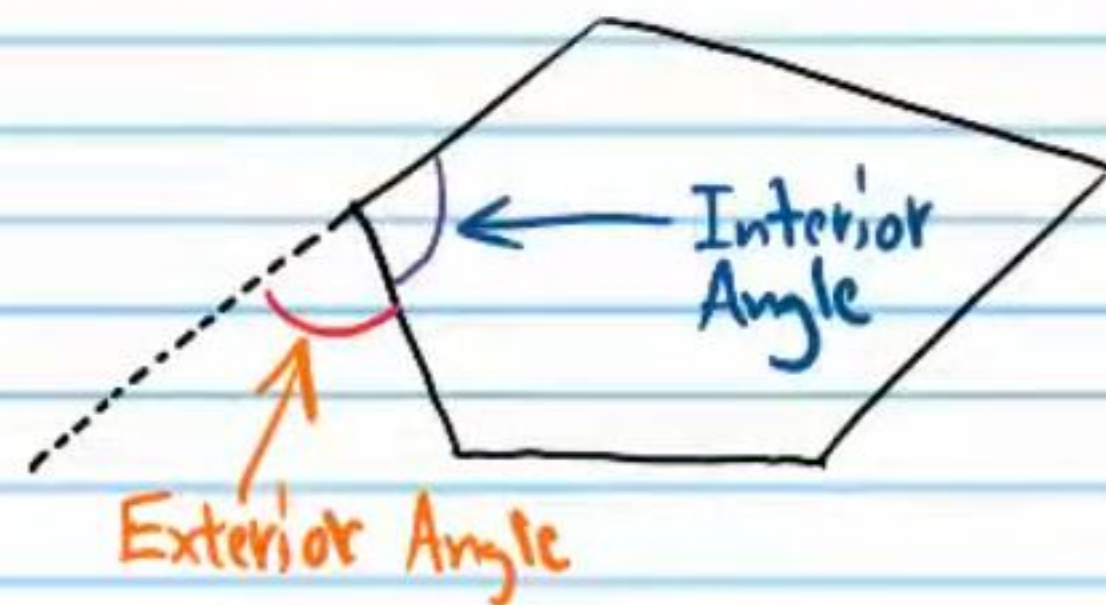
II) $720 = 160 + 130 + 90 + 90 + y + 2y - 70$

$$720 = 400 + 3y$$

$$320 = 3y$$

$$\boxed{y = 106.\bar{6}^\circ}$$

Exterior Angle of a Polygon: An angle whose measure is supplementary to its corresponding interior angle.



Polygon Exterior Angles Theorem: The sum of the measures of the exterior angles of any convex polygon is 360° .

Find the sum of the measures of the exterior angles of the following convex polygons.

- (23) Pentagon (24) Octagon (25) Hexagon (26) Quadrilateral
 360° 360° 360° 360°

Find X in each problem below.

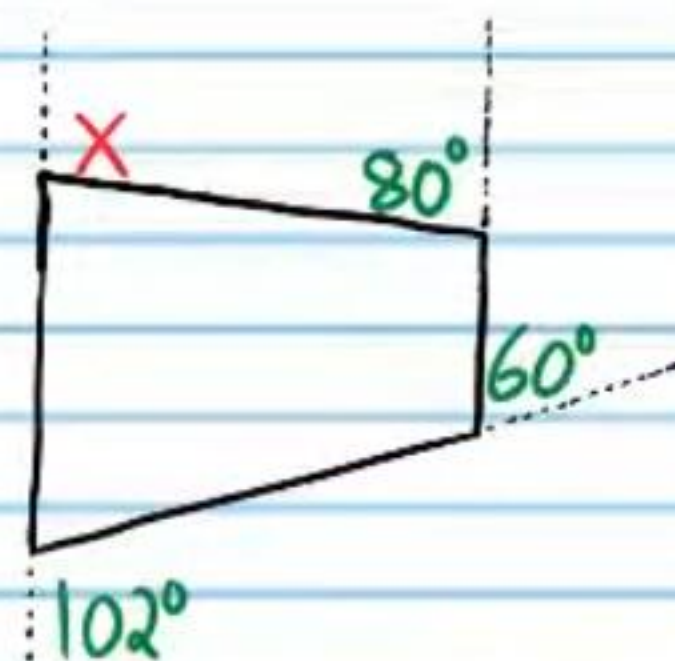


$$360 = 80 + 40 + 90 + 30 + X$$

$$360 = 240 + X$$

$$120^\circ = X$$

(28)

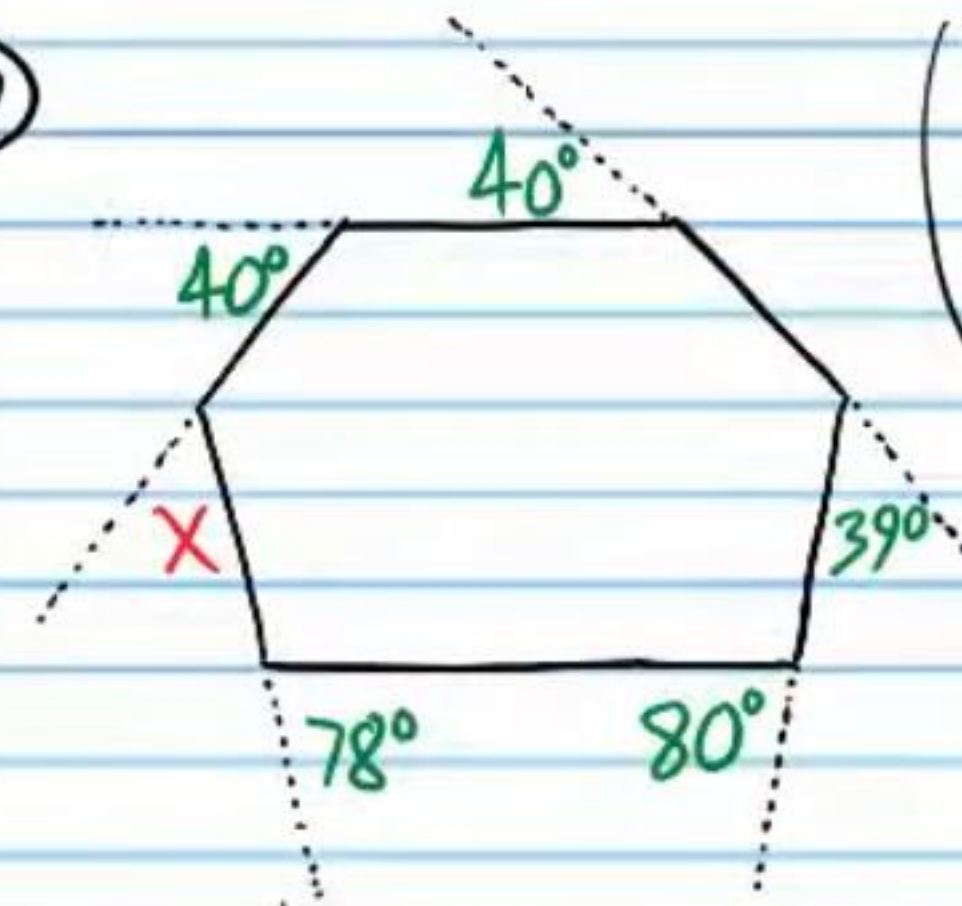


$$360 = X + 80 + 60 + 102$$

$$360 = X + 242$$

$$118^\circ = X$$

(29)

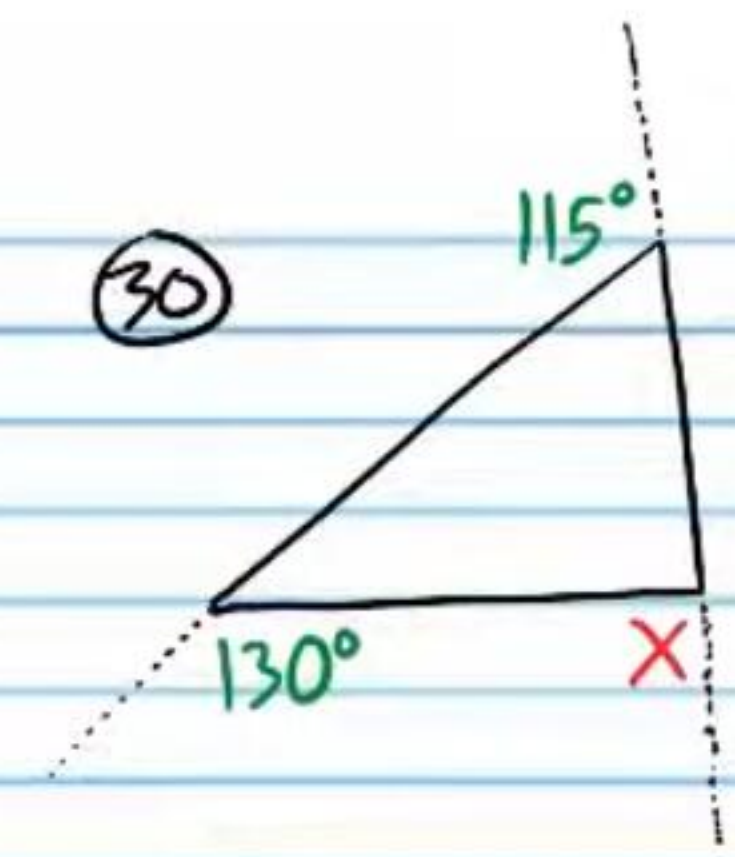


$$360 = 40 + 40 + X + 78 + 80 + 39$$

$$360 = 277 + X$$

$$83^\circ = X$$

(30)



$$360 = 130 + 115 + X$$

$$360 = 245 + X$$

$$115^\circ = X$$

Special Quadrilaterals (Part I)

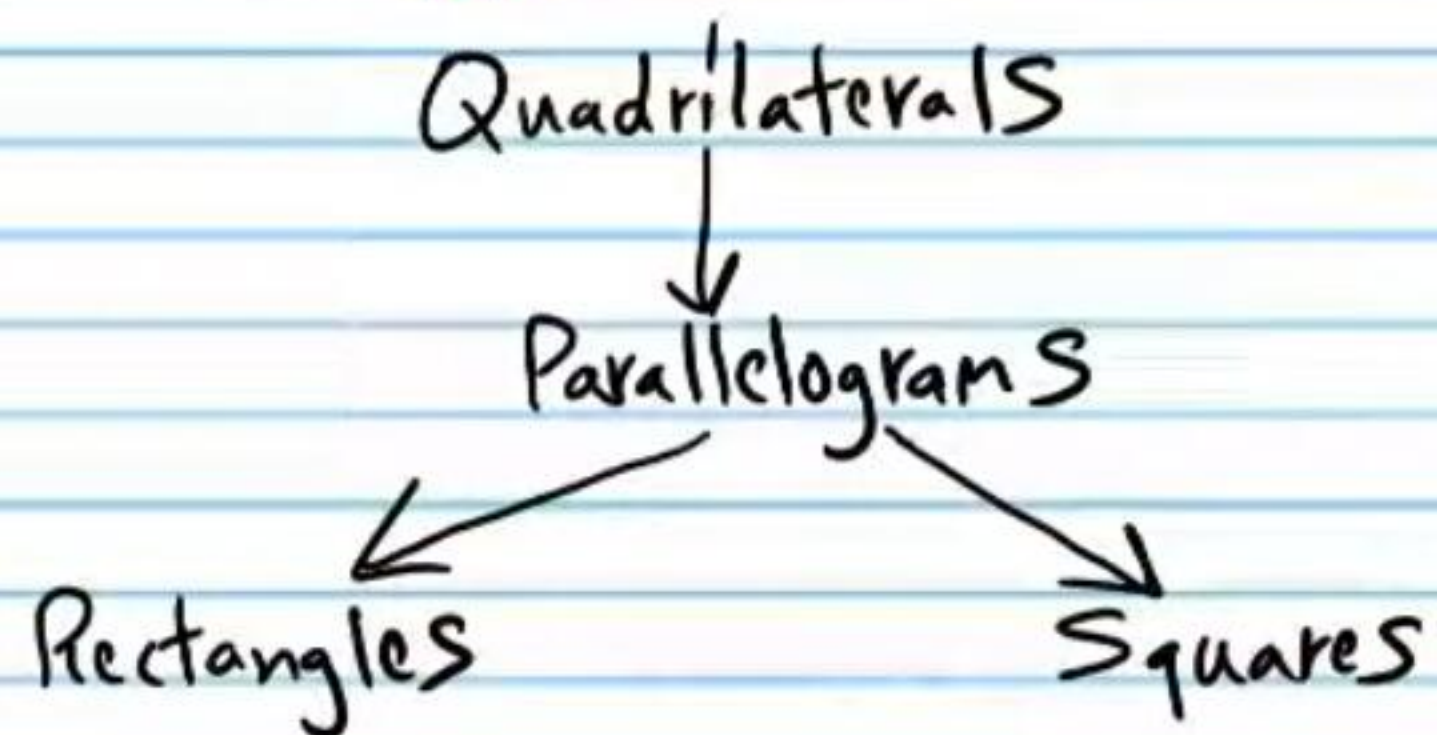
Quadrilateral: A four sided polygon.

Example:

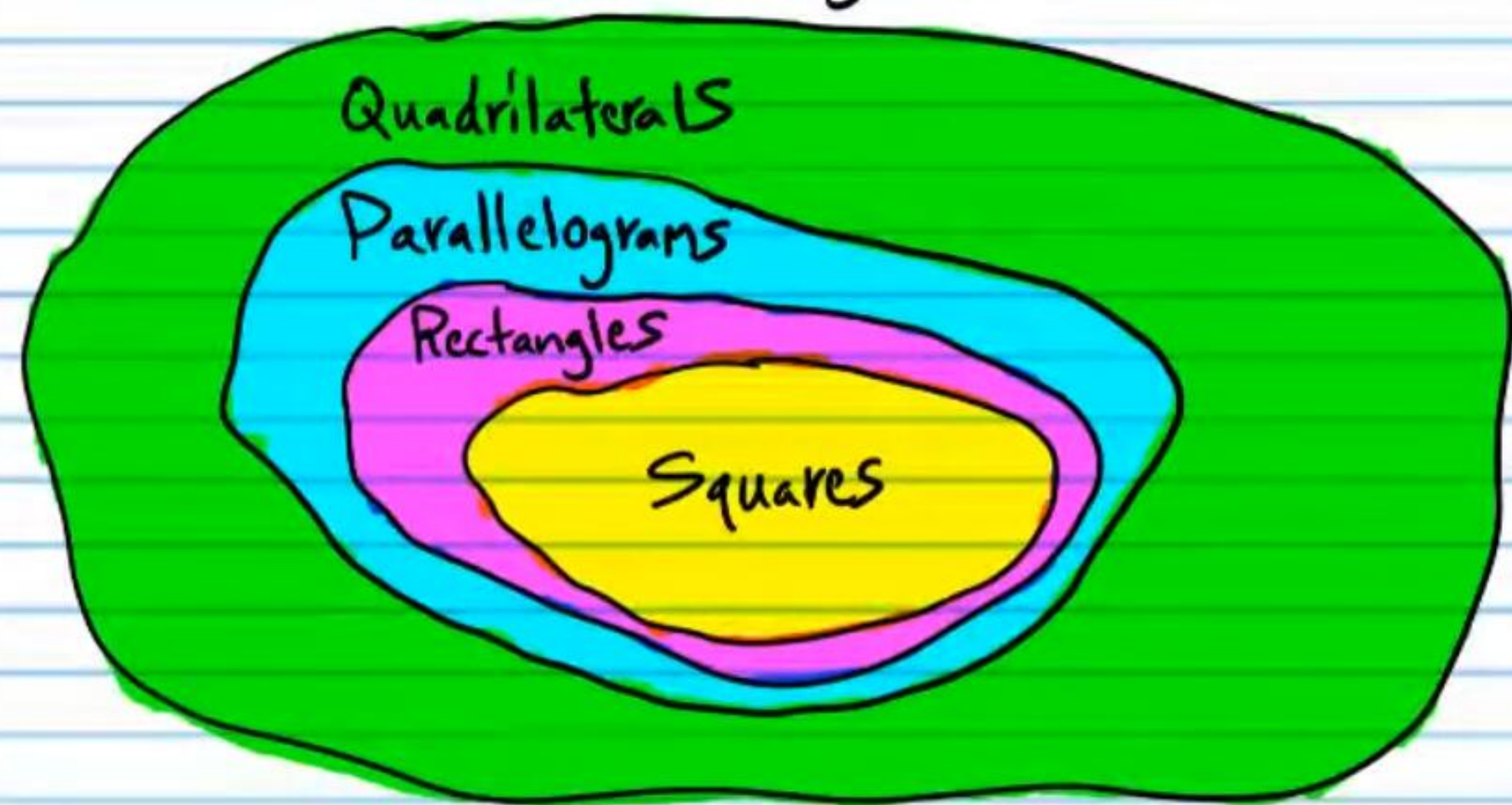


Parallelogram: A quadrilateral with two pairs of parallel sides.

Example:



Venn Diagram

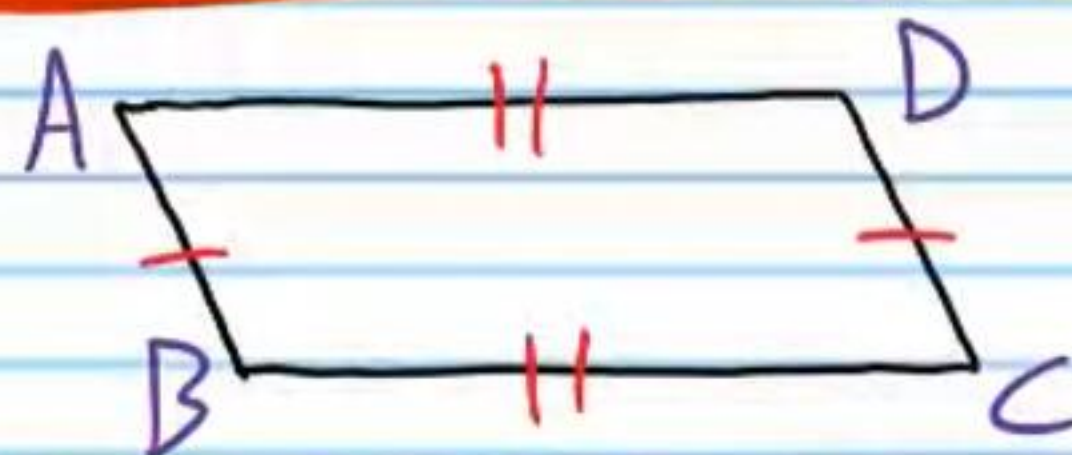


Parallelograms

Parallelogram Opposite Sides Theorem: Opposite sides of parallelograms are congruent.

1) $\overline{AB} \cong \overline{DC}$

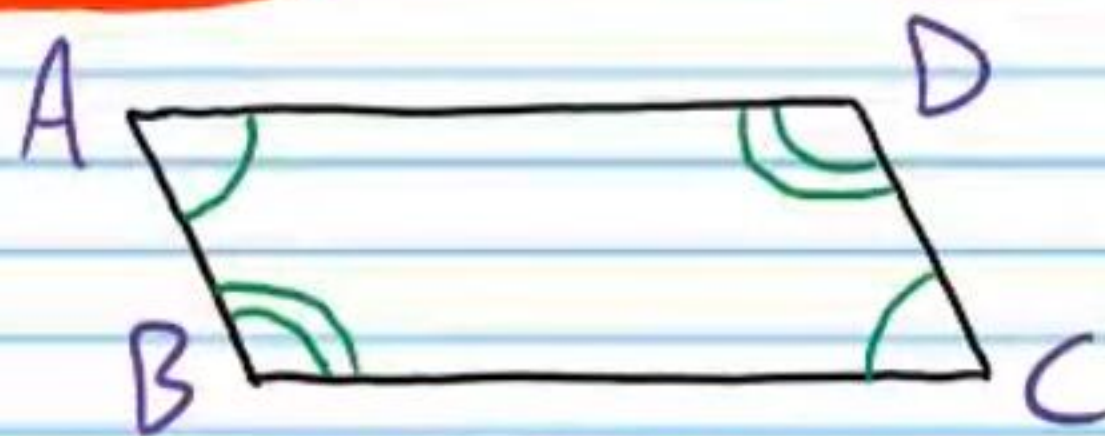
2) $\overline{AD} \cong \overline{BC}$



Parallelogram Opposite Angles Theorem: Opposite angles of parallelograms are congruent.

1) $\angle A \cong \angle C$

2) $\angle B \cong \angle D$



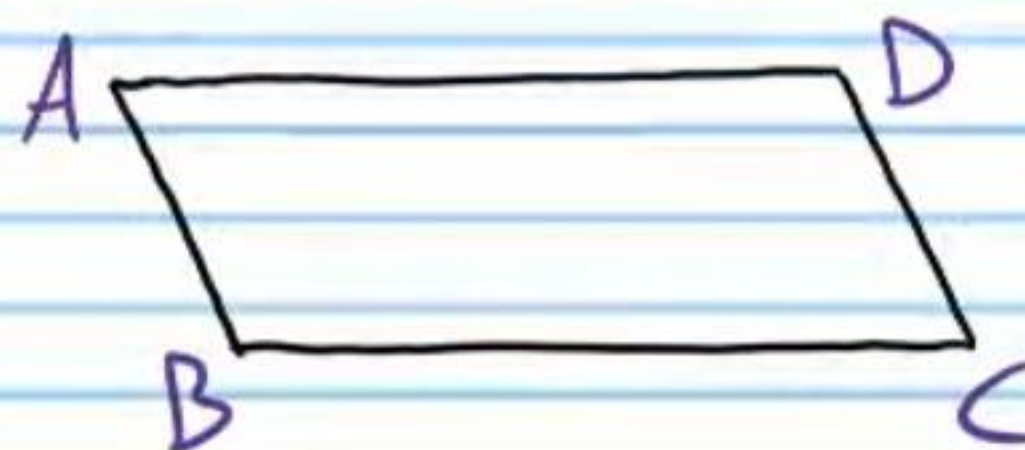
Parallelogram Consecutive Angles Theorem: Consecutive angles of parallelograms are supplementary.

1) $m\angle A + m\angle B = 180^\circ$

2) $m\angle B + m\angle C = 180^\circ$

3) $m\angle C + m\angle D = 180^\circ$

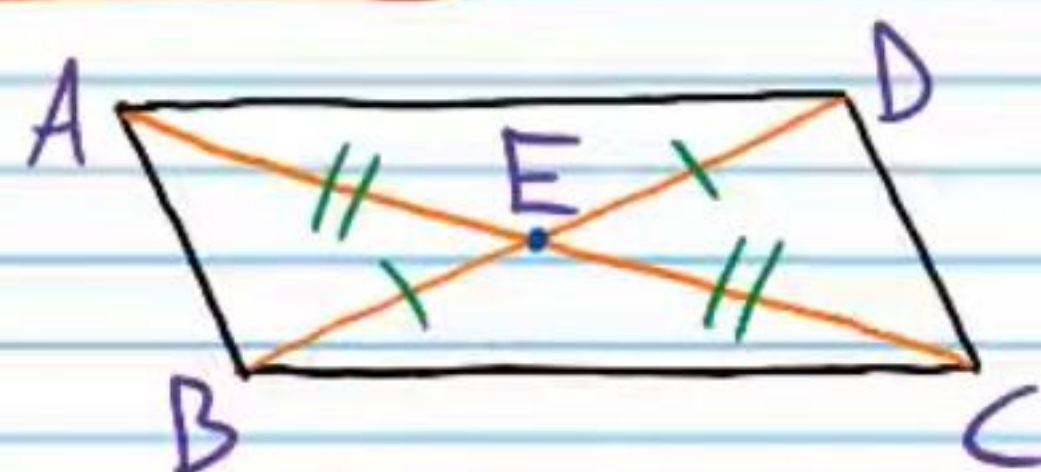
4) $m\angle D + m\angle A = 180^\circ$



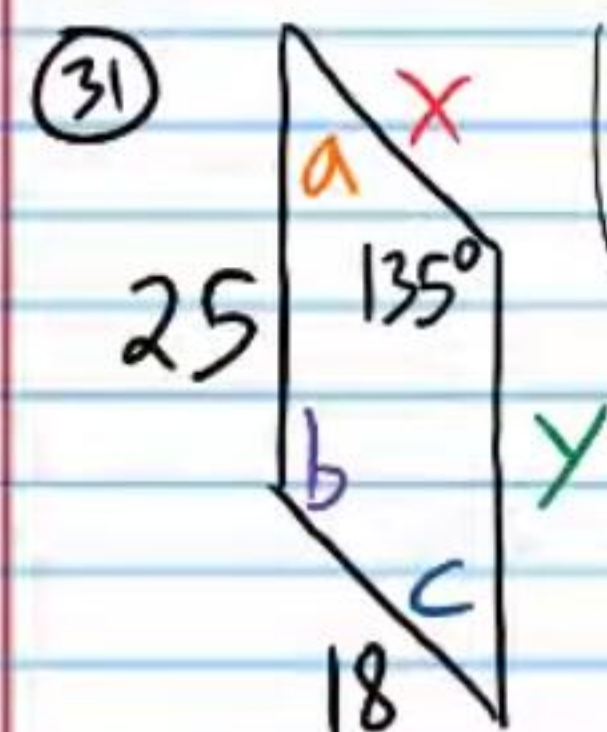
Parallelogram Diagonals Theorem: Diagonals of parallelograms bisect each other.

1) $\overline{BE} \cong \overline{ED}$

2) $\overline{AE} \cong \overline{EC}$



Find the unknown values of the parallelograms below.



$$\begin{aligned} x &= 18 \\ y &= 25 \\ b &= 135^\circ \end{aligned}$$

$$a + 135 = 180$$

$$\boxed{a = 45^\circ}$$

$$\boxed{c = 45^\circ}$$

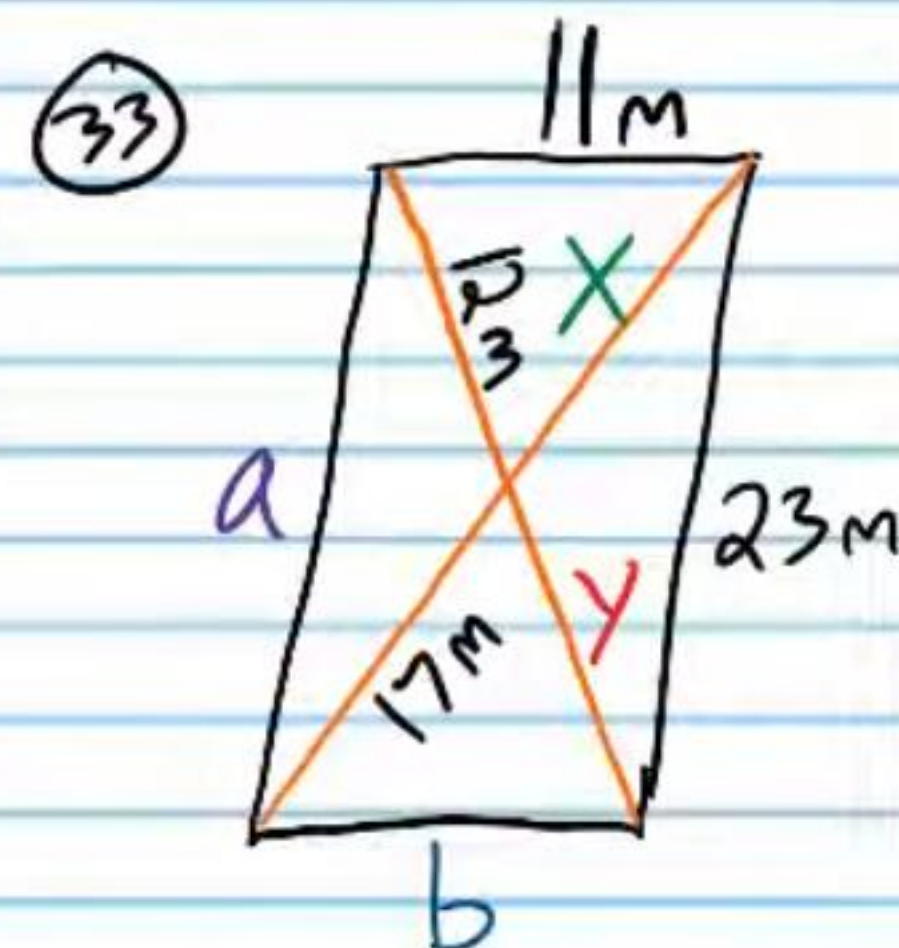


$$\begin{aligned} x &= 21 \\ y &= 15 \\ b &= 130^\circ \end{aligned}$$

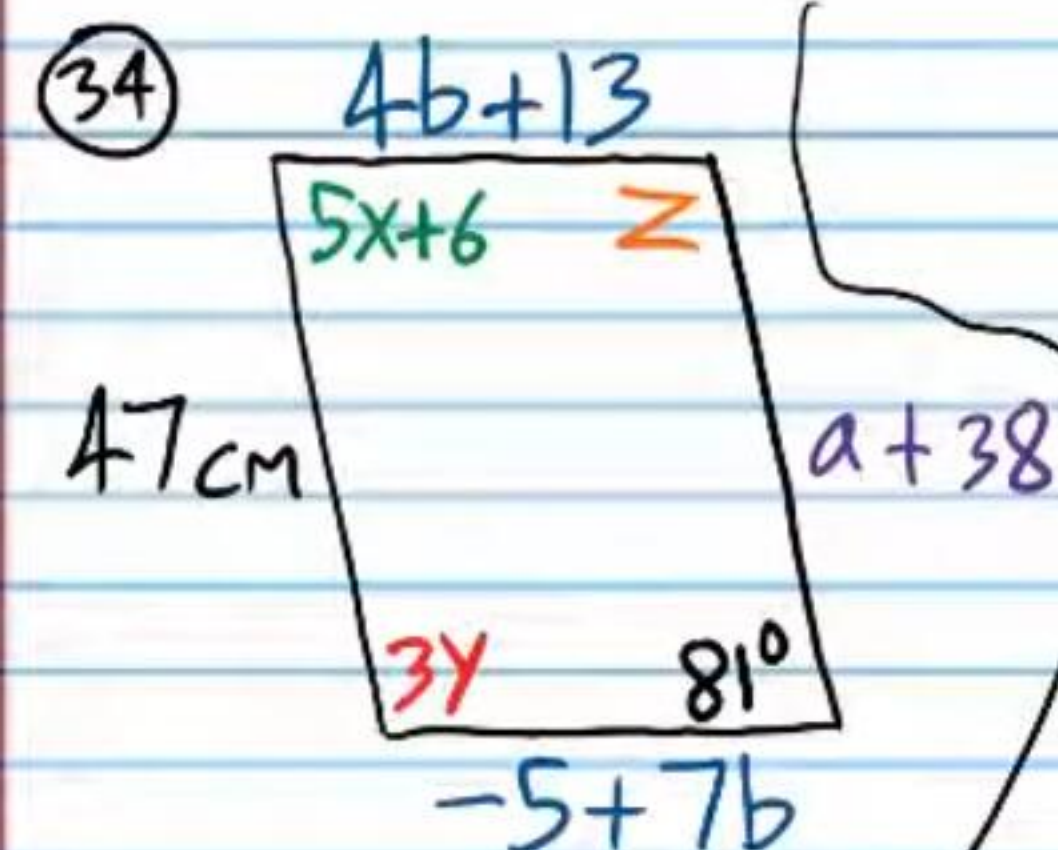
$$180 = 130 + c$$

$$\boxed{50^\circ = c}$$

$$\boxed{a = 50^\circ}$$



$$\begin{aligned} a &= 23m \\ b &= 11m \\ y &= 12m \\ x &= 17m \end{aligned}$$



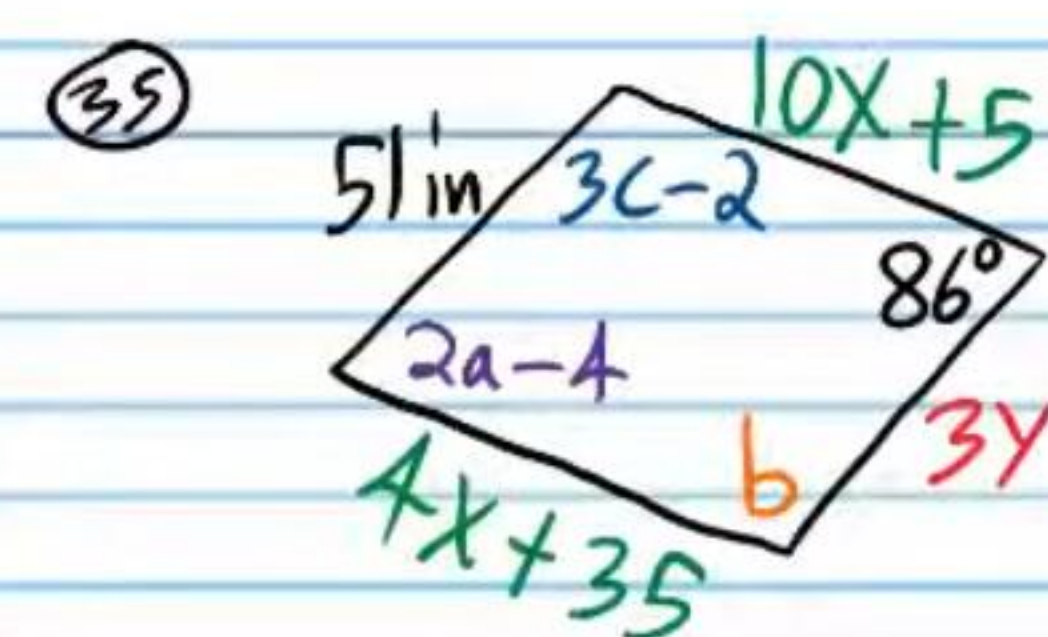
I) $47 = a + 38$
 $\boxed{9 = a}$

II) $4b + 13 = -5 + 7b$
 $18 = 3b$
 $\boxed{6 = b}$

III) $81 = 5x + 6$
 $75 = 5x$
 $\boxed{15 = x}$

IV) $z + 81 = 180$
 $\boxed{z = 99^\circ}$

V) $3y = 99$
 $\boxed{y = 33}$



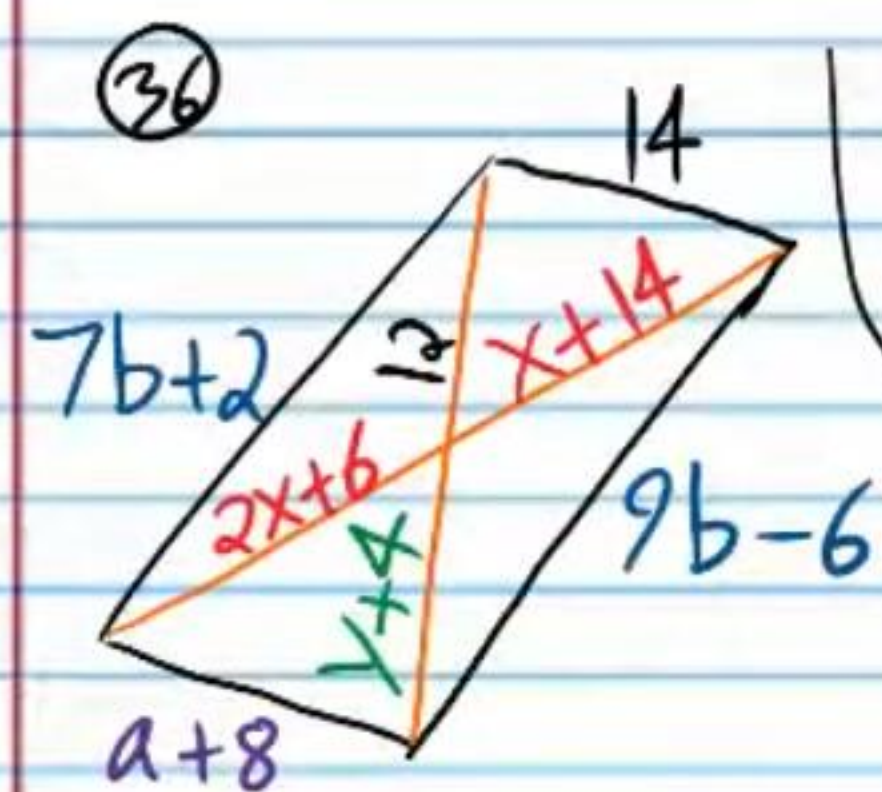
I) $3y = 51$
 $\boxed{y = 17}$

II) $10x + 5 = 4x + 35$
 $6x = 30$
 $\boxed{x = 5}$

III) $86 = 2a - 4$
 $90 = 2a$
 $\boxed{45 = a}$

IV) $b + 86 = 180$
 $\boxed{b = 94^\circ}$

V) $3c - 2 = 94$
 $3c = 96$
 $\boxed{c = 32}$

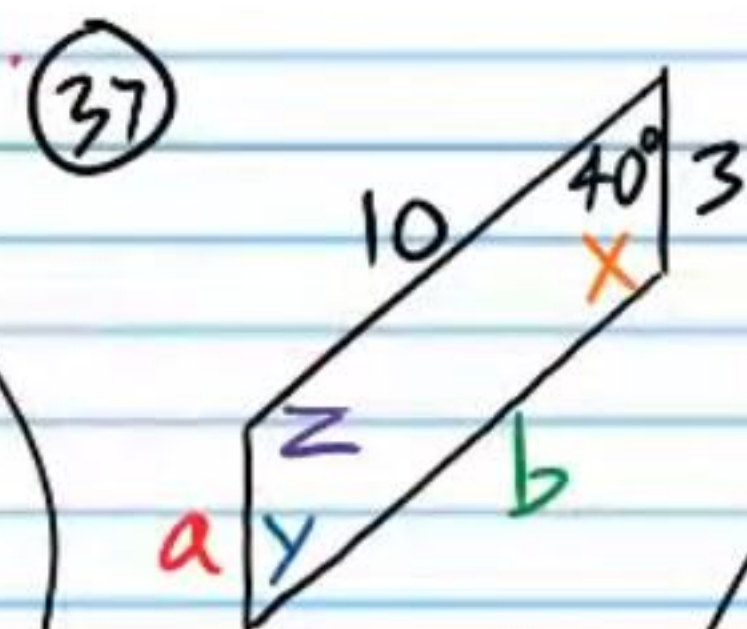


I) $14 = a + 8$
 $\boxed{6 = a}$

II) $7b + 2 = 9b - 6$
 $8 = 2b$
 $\boxed{4 = b}$

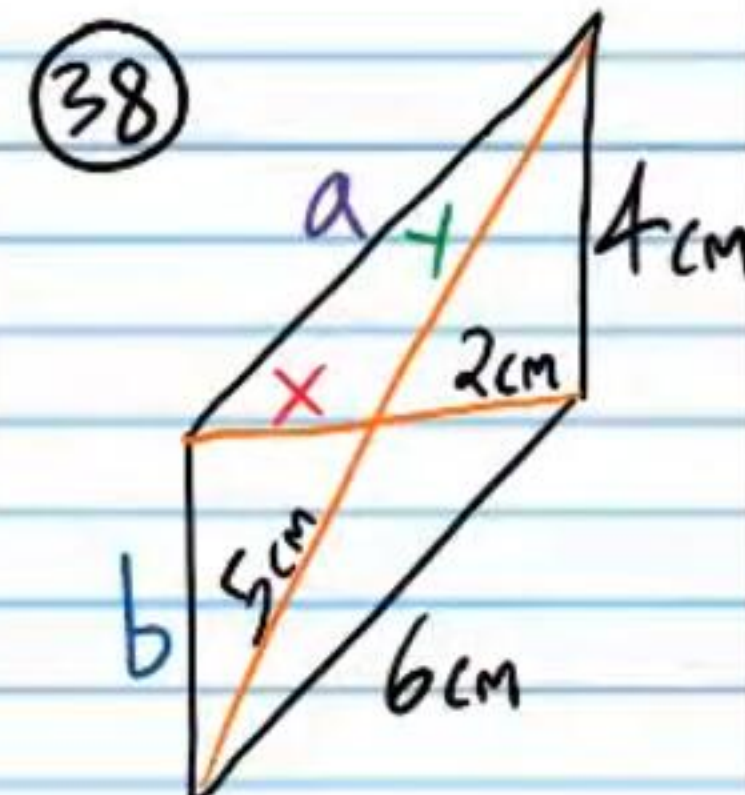
III) $12 = y + 4$
 $\boxed{8 = y}$

IV) $2x + 6 = x + 14$
 $\boxed{x = 8}$



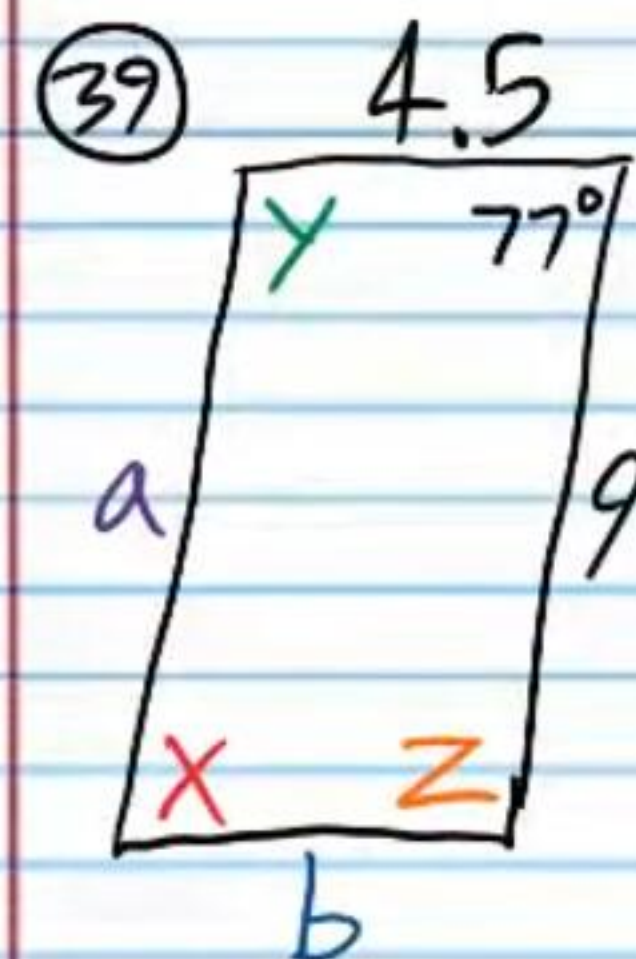
$a = 3$
 $b = 10$
 $y = 40^\circ$

$x + 40 = 180$
 $\boxed{x = 140^\circ}$
 $\boxed{z = 140^\circ}$



$a = 6\text{cm}$
 $b = 4\text{cm}$

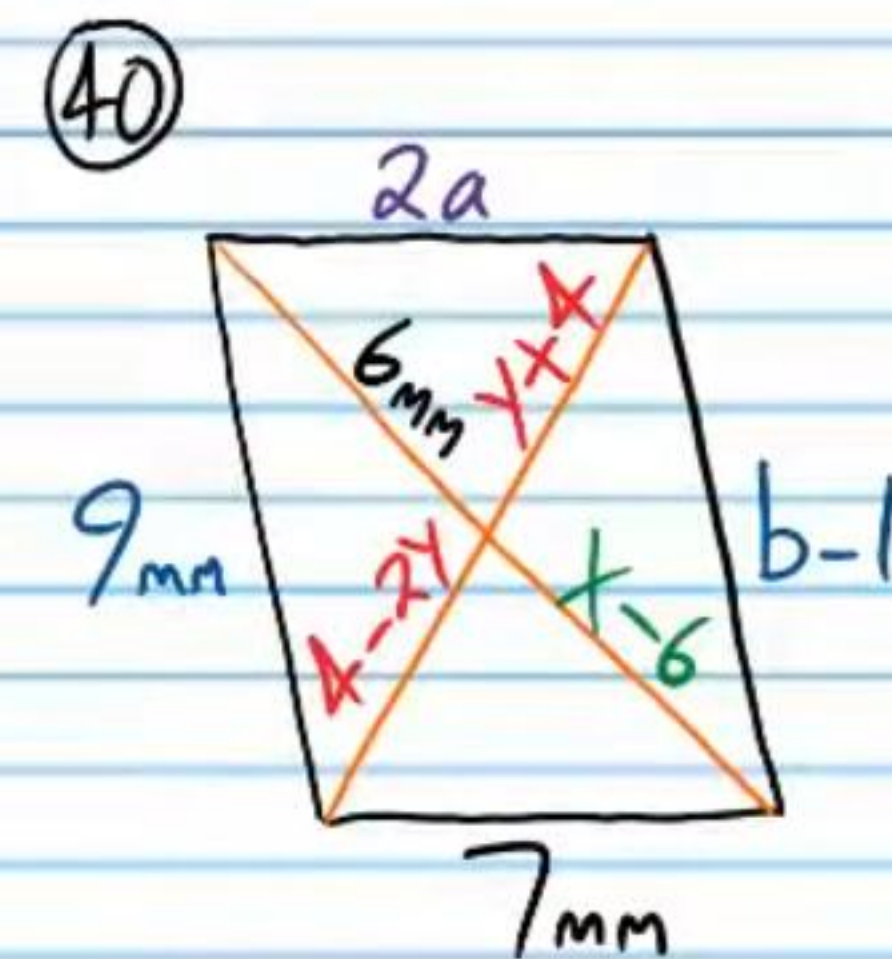
$x = 2\text{cm}$
 $y = 5\text{cm}$



$a = 9$
 $b = 4.5$
 $x = 77^\circ$

$77 + z = 180$
 $\boxed{z = 103^\circ}$

$y = 103^\circ$



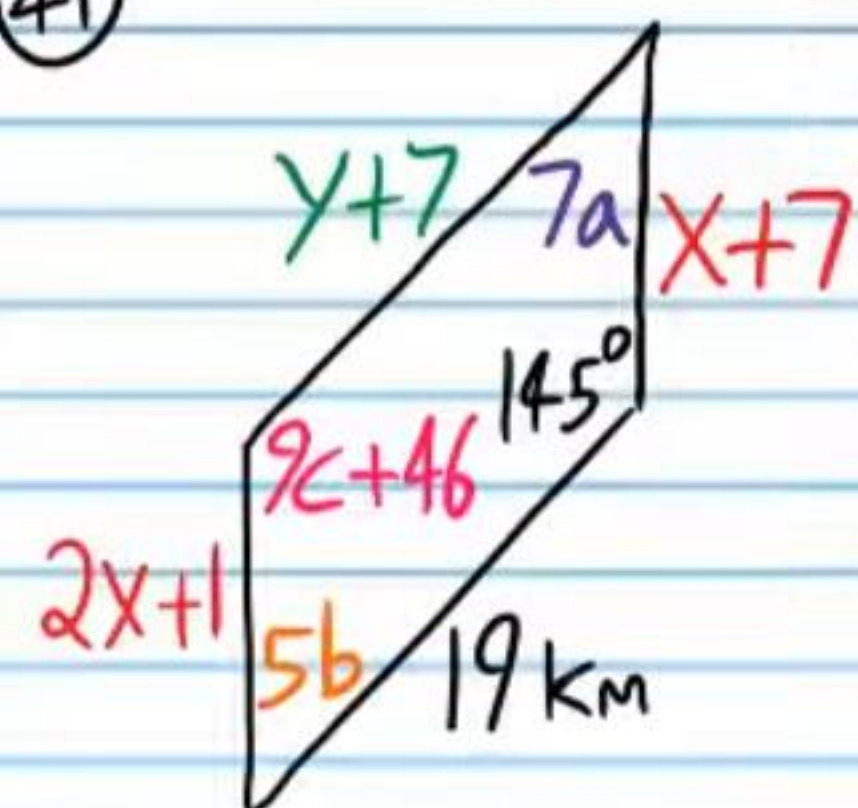
I) $2a = 7$
 $\boxed{a = 3.5}$

II) $9 = b - 1$
 $\boxed{10 = b}$

III) $6 = x - 6$
 $\boxed{12 = x}$

IV) $4 - 2y = y + 4$
 $0 = 3y$
 $\boxed{0 = y}$

41



I) $y+7 = 19$

$y = 12$

II) $2x+1 = x+7$

$x = 6$

III) $145 = 9c+46$

$99 = 9c$

$11 = c$

IV) $5b+145 = 180$

$5b = 35$

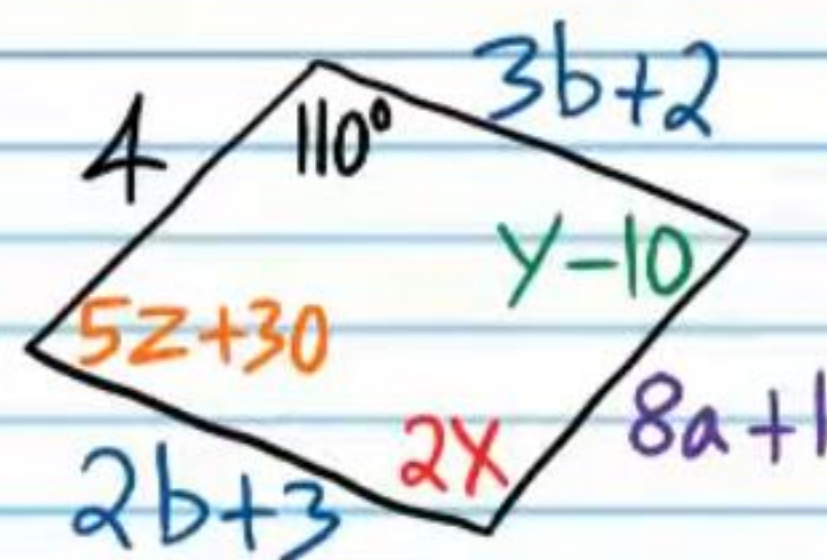
$b = 7$

V) $7a+145 = 180$

$7a = 35$

$a = 5$

42



I) $4 = 8a+1$

$\frac{3}{8} = a$

II) $2b+3 = 3b+2$

$1 = b$

III) $110 = 2x$

$55 = x$

IV) $110 + 5z + 30 = 180$

$140 + 5z = 180$

$5z = 40$

$z = 8$

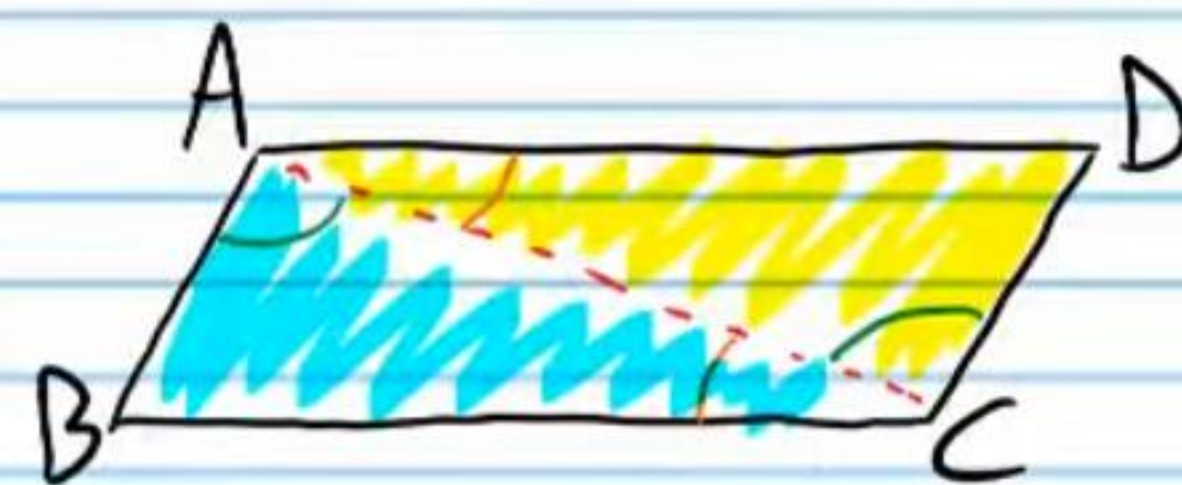
V) $110 + y - 10 = 180$

$100 + y = 180$

$y = 80$

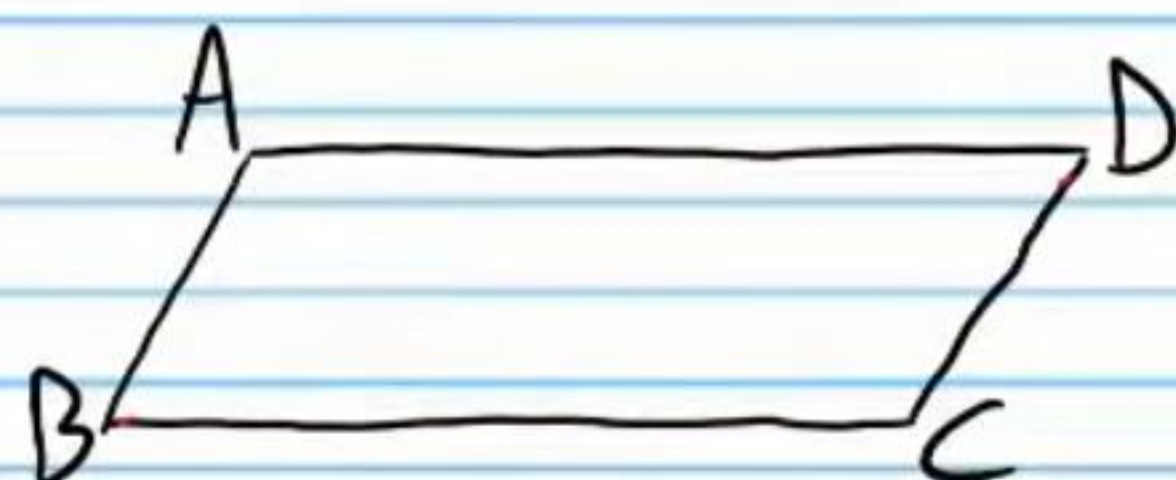
Proving Parallelogram Theorems

43) Prove the Parallelogram Opposite Sides Theorem.



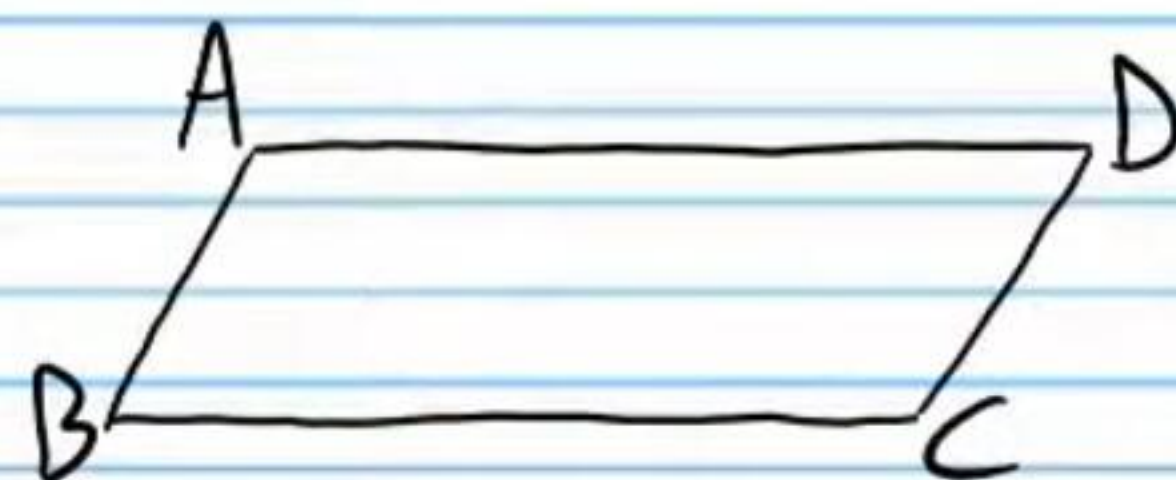
Statement	Reason
I) ABCD is a parallelogram.	Given.
II) $\overline{AB} \parallel \overline{DC}$ & $\overline{AD} \parallel \overline{BC}$.	Def. of $\square$.
III) $\angle DAC \cong \angle BCA$ & $\angle BAC \cong \angle DCA$	Alt. Angles
IV) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$.
V) $\triangle BAC \cong \triangle DCA$	ASA
VI) $\overline{AB} \cong \overline{DC}$ & $\overline{AD} \cong \overline{BC}$	CPCTC

④ Prove the Parallelogram Opposite Angles Theorem.



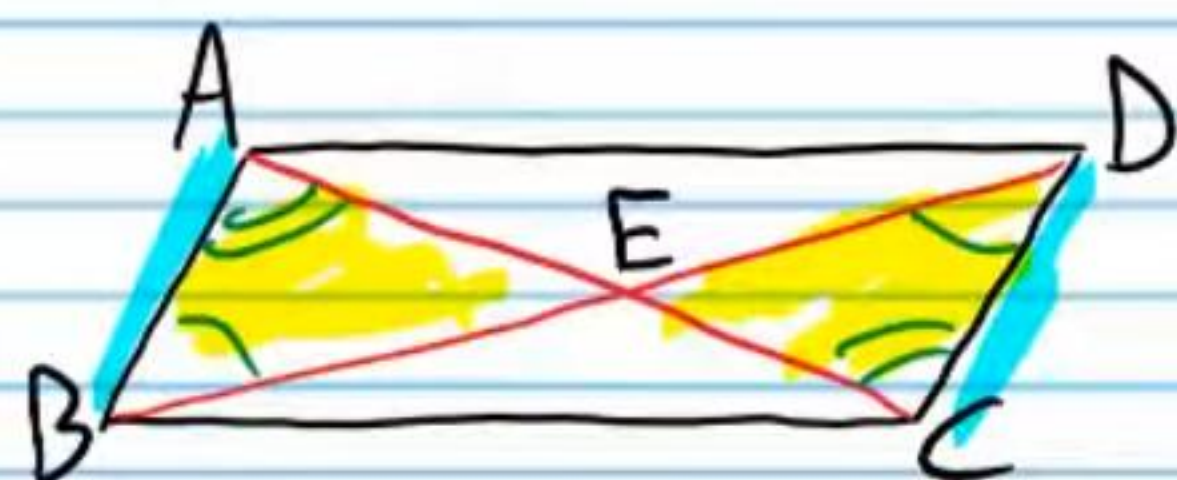
Statement	Reason
I) ABCD is a parallelogram	Given.
II) $\overline{AB} \parallel \overline{DC}$ & $\overline{AD} \parallel \overline{BC}$.	Def. of $\square$
III) $\angle BAC \cong \angle DCA$ & $\angle DAC \cong \angle BCA$	AIAT
IV) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$
V) $\triangle BAC \cong \triangle DCA$	ASA
VI) $\angle B \cong \angle D$	CPCTC
VII) $\angle DBC \cong \angle ADB$ & $\angle ABD \cong \angle BDC$	AIAT
VIII) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$
IX) $\triangle ABD \cong \triangle CDB$	ASA
X) $\angle A \cong \angle C$	CPCTC

⑤ Prove the Parallelogram Consecutive Angles Theorem.



Statement	Reason
I) ABCD is a parallelogram	Given.
II) $\overline{AB} \parallel \overline{DC}$ & $\overline{AD} \parallel \overline{BC}$.	Def. of $\square$.
III) $\angle A$ & $\angle B$ are supplementary	CIAT
IV)	
V)	
VI)	

46) Prove the Parallelogram Diagonals Theorem.



Statement	Reason
I) ABCD is a parallelogram	Given.
II) $\overline{AB} \parallel \overline{DC}$ & $\overline{AD} \parallel \overline{BC}$.	Def. of $\square$.
III) $\angle BAC \cong \angle DCA$ & $\angle ABD \cong \angle CDB$.	AIAT
IV) $\overline{AB} \cong \overline{DC}$	POST
V) $\triangle AEB \cong \triangle CED$.	ASA
VI) $\overline{AE} \cong \overline{EC}$ & $\overline{BE} \cong \overline{ED}$.	CPCTC
VII) $\overline{AC}$ bisects $\overline{BD}$.	Def. of seg bisector.